

First Terminal Examination - 2082

Class : XI

Subject: Biology (Code: 2011)

Full Marks: 75

Time: 3 hrs.

SET 'B'

Pass Marks: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt All Questions

PART : I [BOTANY]

Group 'A' [BOTANY]

Rewrite the best alternative of each question in your answer sheet. [6 × 1 = 6]

1. Group of individuals which resembles with each other very closely & can reproduce to produce their own fertile offspring's is
 - a. Genus
 - b. Family
 - c. Class
 - d. Species
2. The study of fresh water ecosystem & their inhabitants is called
 - a. Lentic ecosystem
 - b. Lotic ecosystem
 - c. Limnology.
 - d. Estuarine
3. Which one of the following is wrong
 - a. Constant unidirectional wind throughout a year causes branching in windward direction that results flag like tree.
 - b. In commensalism, one of the interacting partner is benefited and other partner is neither benefited nor harmed
 - c. Plants growing in strong sunlight are called sciophytes
 - d. Tertiary consumers are also called secondary carnivores
4. ICBN stands for
 - a. International Centre of Botanical Nomenclature
 - b. International Council for Botanical Nomenclature
 - c. International Code of Botanical Nomenclature
 - d. International Code of Binomial Nomenclature

5. The food chain which starts from dead & decaying organic matter is called
 - a. Detritus food chain
 - b. Predatory food chain
 - c. Grazing food chain
 - d. Parasitic food chain
6. In taxonomic hierarchy..... represents lower most rank
 - a. Genus
 - b. Class
 - c. Family
 - d. Species

Group 'B' [BOTANY]

Give short answer to the following questions.

[4 × 4 = 16]

7. Define synecology. Describe the ecological role of temperature on flora and fauna. [1+3]
8. Define Phycology. Write down the general characters of Algae. [1+3]
9. Who proposed five kingdom system of classification? Write down the merits of five kingdom system of classification over two kingdom system. [1+3]
10. Give appropriate scientific term for each of the following. [1×4]
 - a) Group of related species
 - b) Photosynthetic, non-vascular, thalloid & aquatic plants
 - c) Process of formation of soil
 - d) An intraspecific biotic interaction where there is leadership of one followed by other, division of labour and discipline among the members

OR

Distinguish between prokaryotic & eukaryotic cell.

[4]

Group 'C' [BOTANY]

Give long answer to the following questions.

[8 × 2 = 16]

11. Describe the structural & functional aspects of fresh water lentic ecosystem. [4+4]
12. Describe cell structure & scalariform conjugation of *Spirogyra* with necessary illustration. [4+4]

PART : II [ZOOLOGY]

Group 'A' [ZOOLOGY]

Rewrite the best alternative of each question in your answer sheet. [5 × 1 = 5]

13. Study of (molluscan) shells is called
- a. Malacology
 - b. Conchology
 - c. Ethology
 - d. Serology
14. The mutation theory of evolution was proposed by:
- a. Darwin
 - b. Lamarck
 - c. Hugo de Vries
 - d. Huchingson
15. Which one of the following describes correctly the homologous structures?
- a. Organs appearing only in embryonic stage and disappearing later in the adult
 - b. Organs with anatomical similarities but performing different functions
 - c. Organs with anatomical dissimilarities but performing same functions
 - d. Organs that have no function now but had an important function in ancestors
16. Which mechanism introduces new genetic variations in Neo-Darwinism?
- a. Crossing over during meiosis
 - b. Use and disuse of organs
 - c. Environmental pressure
 - d. Lamarckian inheritance
17. The inheritance of acquired character is given by the concept of:
- a. Darwinism
 - b. Lamarckism
 - c. Mendelism
 - d. Neo Lamarckism

Group 'B' [ZOOLOGY]

Give short answers to the following questions.

[4 × 4 =16]

18. Give definition of paleontology. Discuss the various forms of fossils found in nature. [1+3]
19. Describe the experiment that validates the Oparin-Haldane theory with well labelled diagram. [2+2]
20. Briefly Describe the theory of Inheritance. Who proposed this theory? [3+1]
21. Applying Darwinism explain how mosquitoes have become DDT resistant.

Group 'C' [ZOOLOGY]

Give long answers to the following questions.

[2 × 8 = 16]

22. Modern Synthesis Theory is also known as Neo-Darwinism. How do you explain Neo-Darwinism in terms of evolutionary theory? How can you enumerate this theory in your own words? [1+7]
23. How homologous and vestigial organs are supporting the organic evolution of life. Describe with suitable figures in your own words. [4+4]

❧ The End ❧